Semester Project - 2026

IDATT2105 Full-stack application development

Read this document carefully and be sure you take the details into account!

The project assignment is voluntary and can only be carried out by students who receive a C, and want to improve their grade to A or B. Note that all parts of the project (programmed solution, documentation, and presentation) are expected to maintain **very** good quality to achieve grade B or A.

The requirement is to implement a full-stack web application including

- A frontend using Vuejs (v3) related framework and libraries
- A backend using Java (v17, v21 or v25) and Spring boot framework (v3) / Spring framework (v5 (or v6))
- **A database using MySQL (v8 or v9) and/or H2**

The project contains several parts and the more the group does - with good quality - the better! **So, every group should assess what they prioritize and complete the chosen functionalities as much as possible. Unfinished functionality will not be evaluated and should not be part of the delivery!** The evaluation will be based on all parts of project development, such as good code and necessary documentation for testing and further development, also user interface and well-implemented functionality are important in assessment.

Note that plagiarism is not allowed. If the source code of two or more groups is too similar, the grade of all those groups may be adversely affected and can result in being banned from the study!

Note also that the total number of students per group is from **2 to 4**, and the expected effort per student is about 40 hours.

More on general and product requirements are described in the following.

General Requirements

1. The submission deadline is end of week 15 i.e **Friday 10th of April 2026, kl. 14:00**
2. The required materials should be submitted in **Inspera if not otherwise stated in BB.**
3. One submission per group. Remember to write the names of all participants in the group.
4. Technical requirements
    - 4.1. Clean CSS. No use of CSS frameworks like Tailwind
    - 4.2. Spring Boot/REST based backend. Each endpoint should have proper authentication/authorization, for example based on JWT and Spring Security.
    - 4.3. Database communication using plain JDBC or Spring JDBC / JPA
    - 4.4. The solution should contain tests. The code coverage shall be at least 50%
    - 4.5. The group should make sure to use CI/CD during development.
    - 4.6. OWASP and universal design principles/techniques should be used in the project
    - 4.7. Session storage can be used to provide short-lived login session on the frontend
5. Documentations requirements (for each module)

5.1. End points (API) must be documented, for example, using Swagger. Note, an explanation of what the endpoints do and what the different attributes are is required. In addition, code must be documented as usual (javadoc).
5.2. System documentation is also a requirement., i.e., documentation that enables a new developer to quickly get the project up and running for testing and further development (architecture sketches/class diagram). Instructions for how to run the project can preferably be done as a README file, while other documentation should be as a PDF.
5.3. Test data that can be used while testing the app, for example, test user credentials, database credentials, etc.
5.4. The prerequisites must be documented if the project/module is dependent on other projects

6. Submission materials.
   6.1. A zip file including all the modules/files
   6.2. A runnable source code of each project/module. For example
      - Properly documented Source and Test files.
      - Configuration files like pom.xml (maven), package.json, Dockerfile, etc.
      - Flyway or normal DB schema scripts with test data
   6.3. Description of how to run tests and get the system running. This should be made easy (for the user) - as a script, docker/maven command or the like.

Product and requirements

This project focuses on the design and development of a digital internal control system tailored for restaurants, bars, and other food and alcohol serving establishments. The system aims to simplify compliance with health, safety, and alcohol regulations by replacing manual checklists and paper-based routines with a structured digital solution.

The platform will help businesses monitor daily operations, ensure proper documentation, and maintain high standards of hygiene and responsible alcohol service. By centralizing tasks such as temperature logging, cleaning routines, and staff training records, the system improves traceability, reduces human error, and makes it easier to pass inspections from regulatory authorities.

At the architectural level, the system is divided into two independent top-level services: **IK-Mat (food compliance)** and **IK-Alkohol (alcohol compliance)**. These services are separated to reflect different regulatory requirements and operational workflows. However, both services can leverage shared, reusable components under the hood, such as user management, authentication, notifications, document storage, and reporting. This modular approach ensures scalability, maintainability, and efficient development through the reuse of common functionality.

The system should be designed to support **multiple tenants**, allowing multiple organizations to configure and manage their own compliance processes independently within the same platform.

The solution will be user-friendly and accessible across devices, allowing employees and managers to efficiently complete, review, and document required procedures in real time.

The layout and design of user-interface elements such as screens is open-ended and it´s up to each individual group's creativity. Students are also expected to base the system on authentic requirements for IK-Mat and IK-Alkohol, derived from relevant legal authorities and regulations. This includes identifying, interpreting, and translating real-world compliance rules into functional system requirements, ensuring that the solution reflects practical industry standards.

Note: This project is sponsored by the upcoming restaurant *Everest Sushi & Fusion AS* (organization number 937 219 997). Two selected project teams will receive a special gift card from the restaurant. These teams may also be invited to further develop their solutions beyond the course.

Example Features:

- **Digital Checklists:**
  Daily, weekly, and monthly task lists for hygiene, cleaning, and safety procedures.
- **Temperature Logging:**
  Recording and monitoring of food storage temperatures with alerts for deviations.
- **Deviation Management:**
  Ability to report, track, and resolve incidents or non-compliance issues.
- **Alcohol Compliance Tracking:**
  Documentation of routines related to responsible alcohol serving and age verification.
- **User Roles and Access Control:**
  Different permission levels for staff, managers, and administrators.
- **Audit and Inspection Reports:**
  Automatically generated reports for internal reviews and external inspections.
- **Document Export (PDF/JSON):**
  Export of reports, logs, and documentation in PDF for sharing and compliance, or JSON for system integration and data portability.
- **Notifications and Reminders:**
  Alerts for overdue tasks, missing logs, or critical issues.
- **Document Storage:**
  Centralized storage of policies, training materials, and certifications.
- **Mobile Accessibility:**
  Responsive design or mobile app for use on smartphones and tablets.
- **Data Analytics Dashboard:**
  Overview of compliance status, trends, and performance metrics.

Project presentation

The project presentation will be done as a video presentation, delivered as part of the project delivery. More info will come closer to the presentation.

Questions about the assignment?

Use the forum on Blackboard. The idea, however, is that if the desired functionality is ambiguous, the group is largely free to interpret the task themselves. **The group should prioritize the most important functionality and finish that, before taking on more functionality. Therefore, make a list of your prioritizations and why.**